End-Semester Examination December 2024

Name of the Program: B.Tech *Semester: 3rd*

Name of the Course: Electrical Machines-I *Course Code: TEE-302*

Time: 3 Hrs Maximum Marks: 100

Note: -

(i) All questions are compulsory.

(ii) Answer any two sub-questions among a, b & c in each main question

(iii) Total marks in each main question are twenty.

(iv) Each question carries 10 marks

Q1	(20marks)	
(a)	A 500/100V Single phase transformer takes an input of 750VA at no load at a rated voltage. The core loss is 210W. Find (i) The iron loss component of no-load current (ii) The magnetizing component of no load. (iii) No load power factor	CO3
(b)	Explain the open circuit test of the transformer.	
(c)	Discuss the analogies between electrical and magnetic circuits. And explain ferromagnetic material.	
Q2		
(a)	Explain energy and co-energy in rotating machines	
(b)	Explain an electromechanical system and justify why a rotating machine is electromechanical.	
(c)	Explain the following 1. Permeability 2. Magnetic intensity 3. Ampere's law 4. Reluctance	CO2,1
Q3		CO4
(a)	Discuss the No load & load characteristics for separately excited generator.	
(b)	Why are compensating windings employed in a DC machine?	
(c)	What is commutation? Discuss any two commutation methods.	
Q4		CO5
(a)	A 10kW , 260V, 8 pole LAP connected DC generator runs at 1250 rpm. Armature has 500 conductor. Find the useful flux per pole for full load armature ohmic loss of 250W. Take 2.5V as the brush drop at full load.	
(b)	Derive the torque equation of DC motor.	
(c)	Explain the three point starter used for DC motor.	
Q5		CO1,5
(a)	Explain the B-H curve for the magnetic material. Also explain the ohms law for magnetism.	
(b)	A 230 V, 4 pole shunt motor has two circuit armature winding with 500 conductors. The armature circuit resistance is 0.25Ω, field resistance is	

	125Ω and the flux per pole is 0.02Wb. Armature reaction is neglected. If the motor draws 14 A from the mains, then compute a) Speed and total torque developed. b) Shaft power, shaft torque & efficiency with rotational losses equal to 300 watts.	
(c)	Explain the speed torque characteristic of DC motor.	